



Engineering Electromagnetics

工程电磁场

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10: Capacitance



- Using Field Sketches to Estimate Capacitance

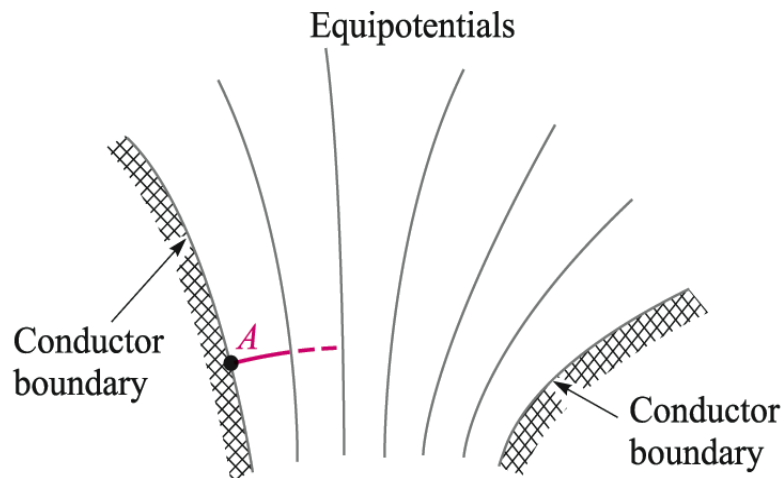
This method employs these properties of conductors and fields:

1. A conductor boundary is an equipotential surface.
2. The electric field intensity and electric flux density are both perpendicular to the equipotential surfaces.
3. $\mathbf{E}$ and $\mathbf{D}$ are therefore perpendicular to the conductor boundaries and possess zero tangential values.
4. The lines of electric flux, or streamlines, begin and terminate on charge and hence, in a charge-free, homogeneous dielectric, begin and terminate only on the conductor boundaries.

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- Using Field Sketches to Estimate Capacitance

Given the conductor boundaries, equipotentials may be sketched in. An attempt is made to establish approximately **equal potential differences** between them.



A line of electric flux density, $\mathbf{D}$, is then started (at point A), and then drawn such that it crosses equipotential lines at right-angles.

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- Using Field Sketches to Estimate Capacitance

Two Lines of $\mathbf{D}$ are shown (sketched in red). All field lines *must* intersect equipotentials at 90° .

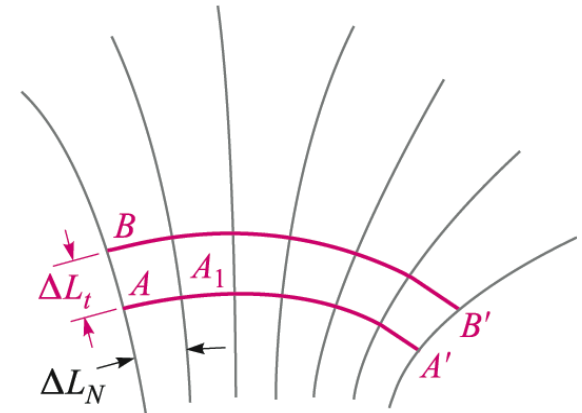
The volume between the two red field lines (having unit depth into the screen) forms a “tube” of flux, of amount $\Delta\psi$. This is the same as the charge on the conductor that is bounded by the tube.

The electric field can now be written as:

$$E = \frac{1}{\epsilon} \frac{\Delta\psi}{\Delta L_t}$$

Then, assuming potential difference ΔV between adjacent equipotentials, we may also write the electric field as:

$$E = \frac{\Delta V}{\Delta L_N}$$



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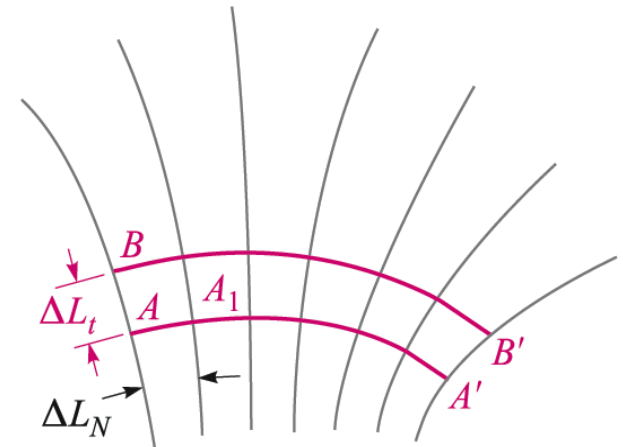
We now have two expressions for electric field, based on the line spacing in the sketch:

$$E = \frac{1}{\epsilon} \frac{\Delta \Psi}{\Delta L_t} \quad E = \frac{\Delta V}{\Delta L_N}$$

Setting the two expressions equal results in

$$\frac{1}{\epsilon} \frac{\Delta \Psi}{\Delta L_t} = \frac{\Delta V}{\Delta L_N}$$

or:
$$\frac{\Delta L_t}{\Delta L_N} = \text{constant} = \frac{1}{\epsilon} \frac{\Delta \Psi}{\Delta V}$$



In other words, the sketch is done such that the ratio $\Delta L_t / \Delta L_N$ is fixed. The easiest way to do this is to make

$$\underline{\Delta L_t = \Delta L_N}$$

So draw the sketch such that *each grid segment is approximately square*

10: Capacitance

• Using Field Sketches to Estimate Capacitance

Here we see the completed sketch, drawn such that the region is divided into **curvilinear squares** 曲边正方形. Each segment along the conductor walls (again of unit depth) contains flux $\Delta\psi$, (or charge ΔQ). Each distance increment between conductors represents a change in potential of ΔV .

The number of segments along the conductor wall is N_Q , the total charge on either conductor is therefore:

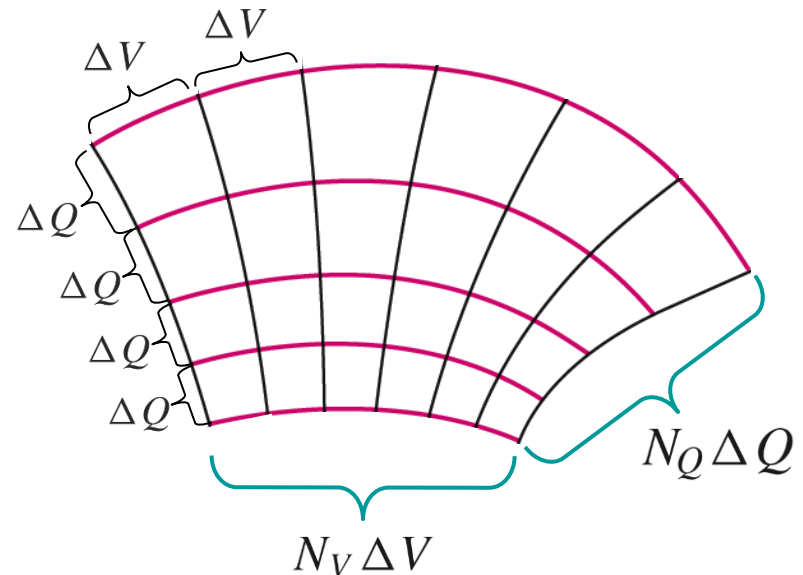
$$Q = N_Q \Delta Q = N_Q \Delta \Psi$$

The number of segments between conductors is N_V , the potential difference between conductor is therefore:

$$V_0 = N_V \Delta V$$

So that the capacitance is now: $C = \frac{N_Q \Delta Q}{N_V \Delta V}$

or finally: $C = \frac{N_Q}{N_V} \epsilon \frac{\Delta L_t}{\Delta L_N} = \epsilon \frac{N_Q}{N_V}$



10: Capacitance

• Using Field Sketches to Estimate Capacitance

Example

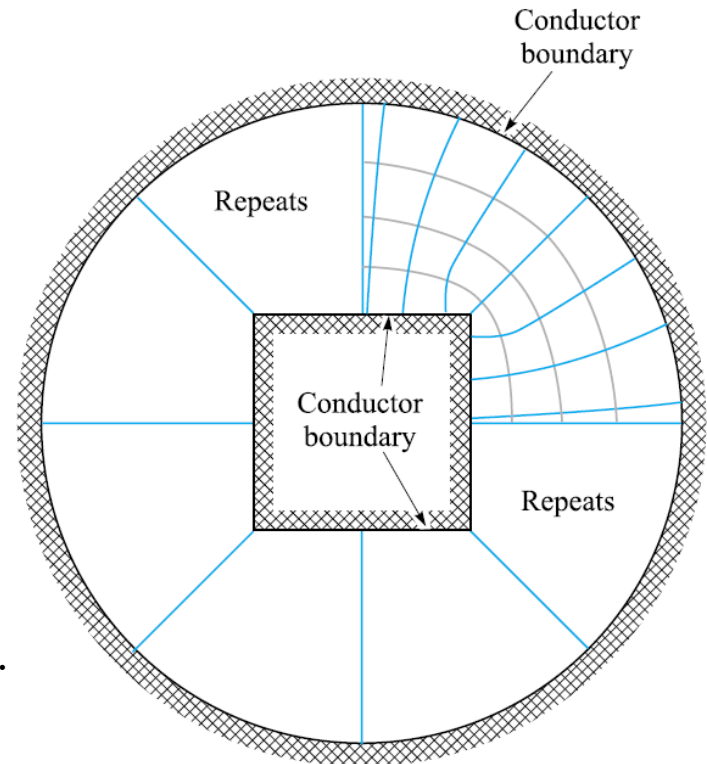
In this transmission line, we wish to estimate the capacitance **per unit length**. The field and equipotential lines in one quadrant are shown.

In this case, division of the range parallel to the conductors into an integral number of squares was not achieved. Instead, over one-eighth of the distance around the perimeter, we have 3.25 divisions.

Between conductors there are exactly four squares.

Therefore:

$$C = \epsilon \frac{N_Q}{N_V} = \epsilon_0 \frac{8 \times 3.25}{4} = \underline{57.6 \text{ pF/m}}$$





10: Capacitance

- Poisson's and Laplace's Equations 泊松、拉普拉斯方程

These **equations** allow one to find the potential field in a region, in which values of potential or electric field are known at its **boundaries**.

Start with **Maxwell's first equation**:

$$\nabla \cdot \mathbf{D} = \rho_v$$

where $\mathbf{D} = \epsilon \mathbf{E}$

and $\mathbf{E} = -\nabla V$

so that

$$\nabla \cdot \mathbf{D} = \nabla \cdot (\epsilon \mathbf{E}) = -\nabla \cdot (\epsilon \nabla V) = \rho_v$$

or finally:

$$\underline{\nabla \cdot \nabla V = -\frac{\rho_v}{\epsilon}}$$

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- Poisson's and Laplace's Equations

Recall the divergence as expressed in rectangular coordinates:

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

...and the gradient:
$$\nabla V = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z$$



Pierre-Simon Laplace
(1749-1827)

then:
$$\begin{aligned} \nabla \cdot \nabla V &= \frac{\partial}{\partial x} \left(\frac{\partial V}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial V}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial V}{\partial z} \right) \\ &= \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} \end{aligned}$$

$\nabla \cdot \nabla$ is abbreviated ∇^2 (and pronounced “del squared”) It is known as the **Laplacian** operator.

10: Capacitance

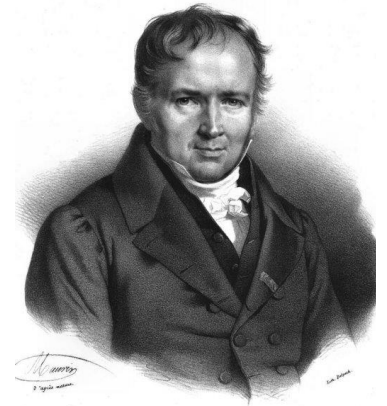


- Poisson's and Laplace's Equations

we already have: $\nabla \cdot \nabla V = -\frac{\rho_v}{\epsilon}$

which becomes:

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = -\frac{\rho_v}{\epsilon}$$



Siméon Denis Poisson
(1781-1840)

This is **Poisson's equation**, as stated in rectangular coordinates.

In the event that there is **zero** volume charge density, the right-hand-side becomes zero, and we obtain **Laplace's equation**:

$$\nabla^2 V = 0$$

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- The Laplacian Operator in the Three Coordinate Systems

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0 \quad (\text{rectangular}) \quad (\text{Laplace's equation})$$

$$\nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \left(\frac{\partial^2 V}{\partial \phi^2} \right) + \frac{\partial^2 V}{\partial z^2} \quad (\text{cylindrical})$$

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} \quad (\text{spherical})$$

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- Example: Parallel Plate Capacitor

In this case, the plate separation, d , is assumed much less than the smallest plate dimension. Therefore V can be assumed to vary only with x .

Laplace's equation reduces to:

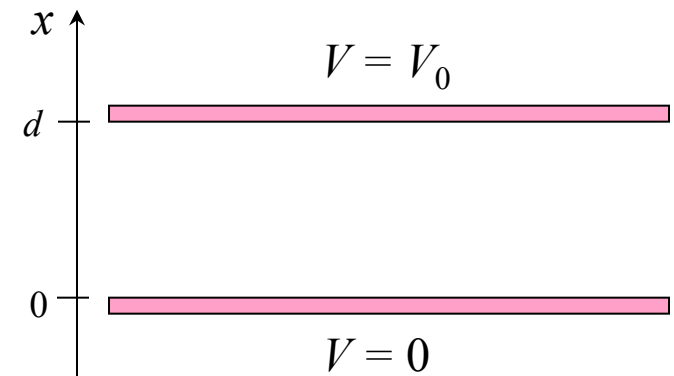
$$\frac{d^2 V}{dx^2} = 0$$

Integrating once, obtain:

$$\frac{dV}{dx} = A$$

Integrate a second time to get: $V = Ax + B$

where A and B are **integration constants** that are to be evaluated subject to the boundary conditions.



Boundary conditions:

1. $V = 0$ at $x = 0$
2. $V = V_0$ at $x = d$

10: Capacitance



- Finding Interior Potential Field

We now have: $V = Ax + B$

Apply boundary condition 1:

$$0 = A(0) + B \longrightarrow \underline{B = 0}$$

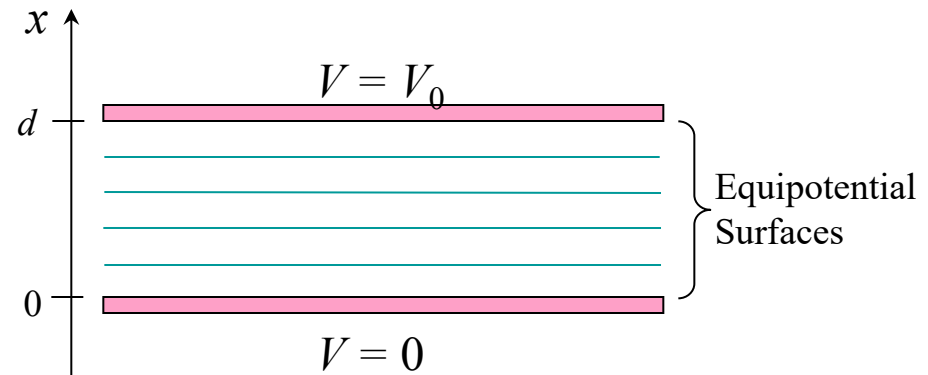
Apply boundary condition 2:

$$V_0 = Ad \longrightarrow \underline{A = \frac{V_0}{d}}$$

Finally:

$$V = \frac{V_0 x}{d}$$

This function is pictured by the equipotential surfaces shown in the capacitor, in which there is a constant voltage difference between adjacent surfaces



Boundary conditions:

1. $V = 0$ at $x = 0$
2. $V = V_0$ at $x = d$

10: Capacitance

- Finding the Electric Field, Charge, and Capacitance

Start with: $V = V_0 \frac{x}{d}$

Then: $\mathbf{E} = -\nabla V = -\frac{V_0}{d} \mathbf{a}_x$

From which: $\mathbf{D} = -\epsilon \frac{V_0}{d} \mathbf{a}_x$

At the lower plate surface ($x = 0$):

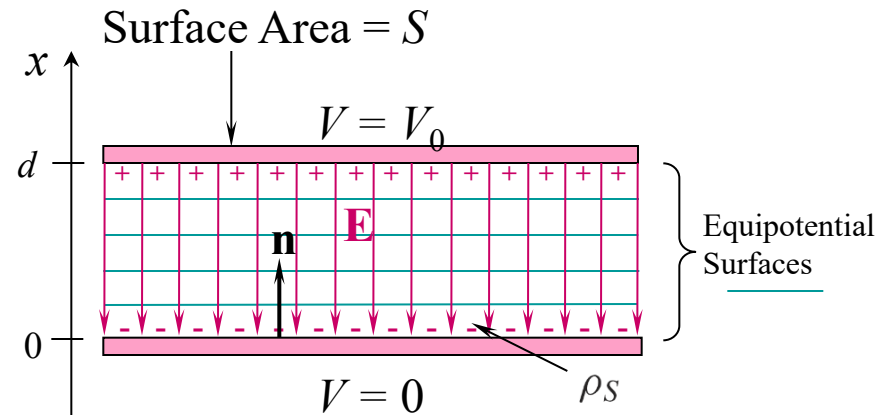
$$\mathbf{D}_S = \mathbf{D}|_{x=0} = -\epsilon \frac{V_0}{d} \mathbf{a}_x \quad \text{and} \quad \mathbf{n} = \mathbf{a}_x$$

Then use: $\mathbf{D} \cdot \mathbf{n}|_S = \rho_S$ to find $D_N = -\epsilon \frac{V_0}{d} = \rho_S$

The charge on the lower plate is now:

$$Q = \int_S \frac{-\epsilon V_0}{d} dS = -\epsilon \frac{V_0 S}{d} \quad \text{and the capacitance is}$$

$$C = \frac{|Q|}{V_0} = \frac{\epsilon S}{d}$$



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• Another Example: Coaxial Transmission Line

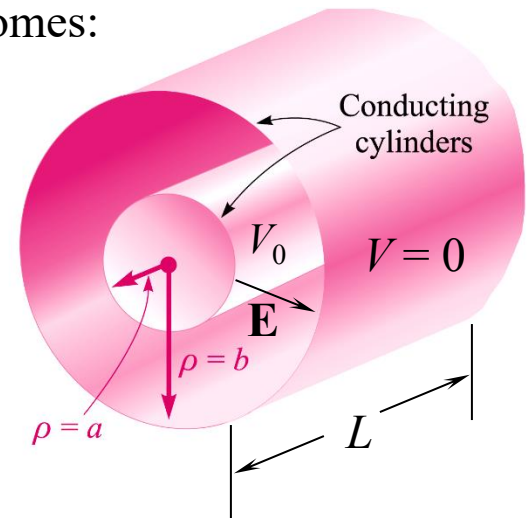
As V is assumed to vary with radius only, Laplace's equation becomes:

$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) = 0 \quad (\text{not valid at } \rho = 0)$$

Our goal is to evaluate the potential function in the region
($a < \rho < b$)

Integrate once: $\rho \frac{dV}{d\rho} = A$

.. and a second time: $V = A \ln \rho + B$



Boundary conditions:

1. $V = 0$ at $\rho = b$
2. $V = V_0$ at $\rho = a$

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- Another Example: Coaxial Transmission Line

Now have: $V = A \ln \rho + B$

Apply boundary condition 1:

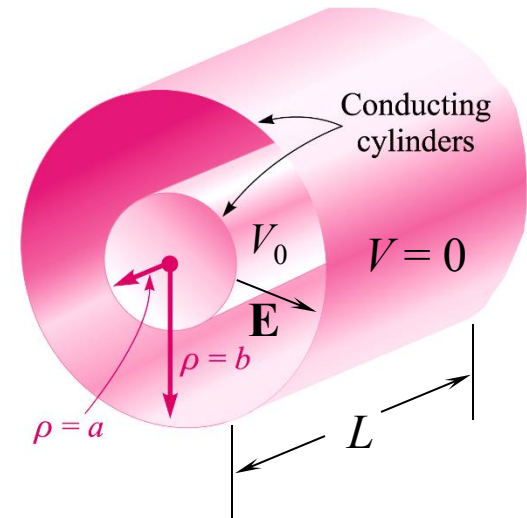
$$0 = A \ln(b) + B \Rightarrow \underline{B = -A \ln(b)}$$

Apply boundary condition 2:

$$V_0 = A \ln(a) - A \ln(b) = A \ln(a/b) \Rightarrow \underline{A = -\frac{V_0}{\ln(b/a)}}$$

Putting it all together:

$$V(\rho) = -\frac{V_0}{\ln(b/a)} [\ln(\rho) - \ln(b)] = \boxed{V_0 \frac{\ln(b/\rho)}{\ln(b/a)}}$$



Boundary conditions:

1. $V = 0$ at $\rho = b$
2. $V = V_0$ at $\rho = a$

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- Another Example: Coaxial Transmission Line

We have found the potential field between conductors: $V(\rho) = V_0 \frac{\ln(b/\rho)}{\ln(b/a)}$

Then:

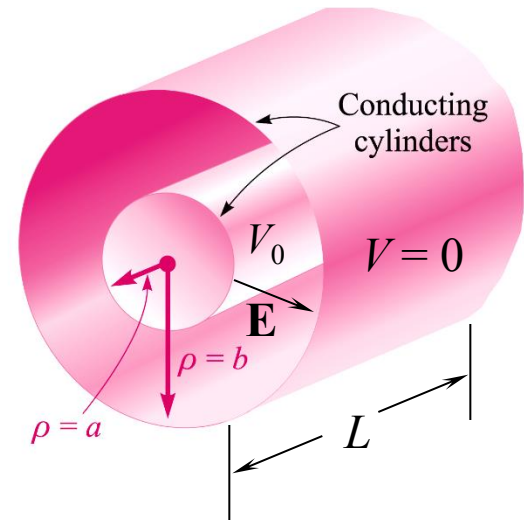
$$\mathbf{E} = -\nabla V = -\frac{dV}{d\rho} \mathbf{a}_\rho = \frac{V_0}{\rho} \frac{1}{\ln(b/a)} \mathbf{a}_\rho$$

The charge density on the inner conductor is:

$$\rho_s = \mathbf{D} \cdot \mathbf{a}_\rho \Big|_{\rho=a} = \frac{\epsilon V_0}{a} \frac{1}{\ln(b/a)} \text{ C/m}^2$$

The total charge on the inner conductor is:

$$Q = \int_S \rho_s da = 2\pi a L \rho_s = \frac{2\pi \epsilon L V_0}{\ln(b/a)} \text{ C}$$



...and the capacitance is finally:

$$C = \frac{Q}{V_0} = \frac{2\pi \epsilon L}{\ln(b/a)} \text{ F}$$

10: Capacitance

• Another Example: Angled Plate Geometry

Cylindrical coordinates are used, **with the assumption that potential varies only with ϕ .**

Laplace's equation becomes:

$$\nabla^2 V = \frac{1}{\rho^2} \frac{d^2 V}{d\phi^2} = 0 \quad (\rho > 0)$$

Integrate once: $\frac{dV}{d\phi} = A$

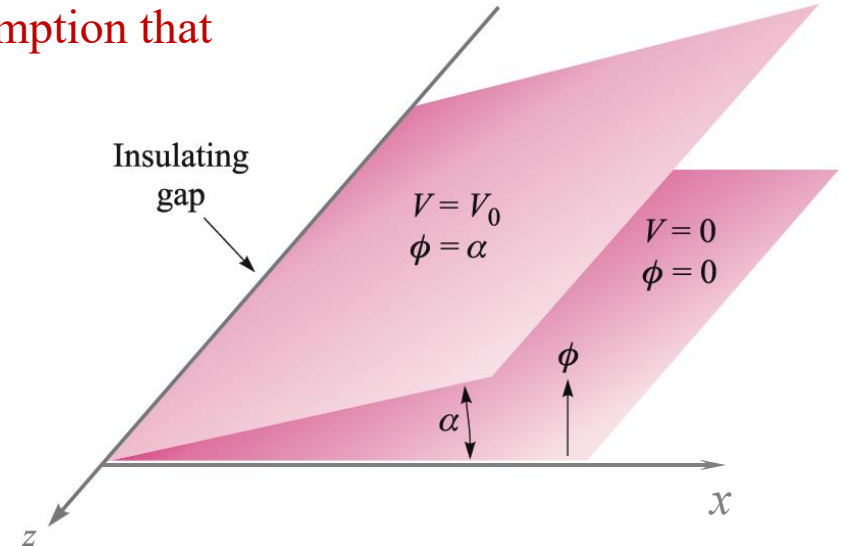
...and integrate again: $V(\phi) = A\phi + B$

First boundary condition: $0 = A(0) + B \Rightarrow B = 0$

Second boundary condition: $V_0 = A\alpha \Rightarrow A = \frac{V_0}{\alpha}$

Finally: $V(\phi) = V_0 \frac{\phi}{\alpha}$

Then: $\mathbf{E} = -\nabla V = -\frac{1}{\rho} \frac{dV}{d\phi} \mathbf{a}_\phi = -\frac{V_0}{\alpha\rho} \mathbf{a}_\phi$

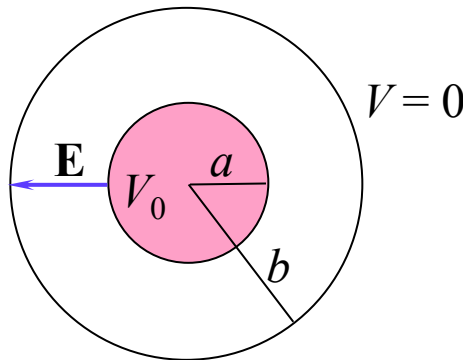


Boundary Conditions:

1. $V=0$ at $\phi = 0$
2. $V=V_0$ at $\phi = \alpha$

10: Capacitance

- Another Example: Concentric Sphere Geometry



Boundary Conditions:

1. $V = 0$ at $r = b$
2. $V = V_0$ at $r = a$

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} \quad (\text{spherical})$$

10: Capacitance



- Another Example: Concentric Sphere Geometry

V is assumed to vary only with radius. Laplace's equation then becomes:

$$\nabla^2 V = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dV}{dr} \right) = 0 \quad \text{or:} \quad \frac{d}{dr} \left(r^2 \frac{dV}{dr} \right) = 0$$

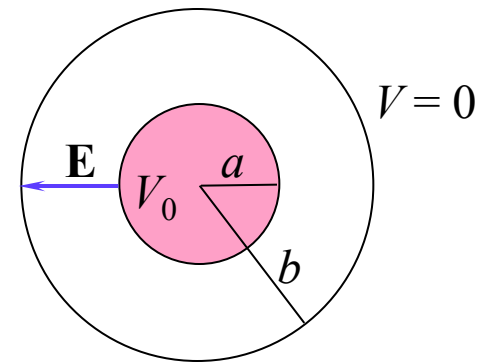
Integrate once: $\frac{dV}{dr} = \frac{A}{r^2}$..and again: $V(r) = -\frac{A}{r} + B$

Boundary condition 1: $0 = -\frac{A}{b} + B \Rightarrow \underline{B = \frac{A}{b}}$

Boundary condition 2: $V_0 = -\frac{A}{a} + \frac{A}{b} \Rightarrow \underline{A = \frac{V_0}{(1/b) - (1/a)}}$

Finally:

$$V(r) = V_0 \frac{(1/r) - (1/b)}{(1/a) - (1/b)}$$



Boundary Conditions:

1. $V = 0$ at $r = b$
2. $V = V_0$ at $r = a$

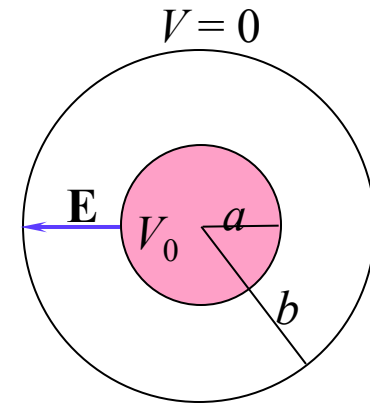
10: Capacitance

- Another Example: Concentric Sphere Geometry

We now have: $V(r) = V_0 \frac{(1/r) - (1/b)}{(1/a) - (1/b)} \quad (a < r < b)$

Electric field between conductors:

$$\mathbf{E} = -\nabla V = \frac{dV}{dr} \mathbf{a}_r = \frac{V_0}{r^2 [(1/a) - (1/b)]} \mathbf{a}_r \quad \text{V/m}$$



Charge density on inner conductor:

$$\rho_s = \mathbf{D} \cdot \mathbf{a}_r \Big|_{r=a} = \frac{\epsilon V_0}{a^2 [(1/a) - (1/b)]} \quad \text{C/m}^2$$

Total charge on inner conductor:

$$Q = \int_S \rho_s da = 4\pi a^2 \rho_s = \frac{4\pi\epsilon V_0}{[(1/a) - (1/b)]} \quad \text{C}$$

Capacitance: $C = \frac{Q}{V_0} = \frac{4\pi\epsilon}{[(1/a) - (1/b)]} \quad \text{F}$

Given the potential field $V = (A\rho^4 + B\rho^{-4}) \sin 4\phi$: (a) Show that $\nabla^2 V = 0$.
(b) Select A and B so that $V = 100$ V and $|\mathbf{E}| = 500$ V/m at $P(\rho = 1, \phi = 22.5^\circ, z = 2)$.

$$\nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \left(\frac{\partial^2 V}{\partial \phi^2} \right) + \frac{\partial^2 V}{\partial z^2} \quad (\text{cylindrical})$$

$$\nabla V = \frac{\partial V}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V}{\partial z} \mathbf{a}_z \quad (\text{cylindrical})$$

Given the potential field $V = (A\rho^4 + B\rho^{-4}) \sin 4\phi$:

a) Show that $\nabla^2 V = 0$: In cylindrical coordinates,

$$\begin{aligned}\nabla^2 V &= \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} \\&= \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho(4A\rho^3 - 4B\rho^{-5}) \right) \sin 4\phi - \frac{1}{\rho^2} 16(A\rho^4 + B\rho^{-4}) \sin 4\phi \\&= \frac{16}{\rho} (A\rho^3 + B\rho^{-5}) \sin 4\phi - \frac{16}{\rho^2} (A\rho^4 + B\rho^{-4}) \sin 4\phi = 0\end{aligned}$$

b) Select A and B so that $V = 100$ V and $|\mathbf{E}| = 500$ V/m at $P(\rho = 1, \phi = 22.5^\circ, z = 2)$: First,

$$\begin{aligned}\mathbf{E} &= -\nabla V = -\frac{\partial V}{\partial \rho} \mathbf{a}_\rho - \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi \\&= -4 \left[(A\rho^3 - B\rho^{-5}) \sin 4\phi \mathbf{a}_\rho + (A\rho^3 + B\rho^{-5}) \cos 4\phi \mathbf{a}_\phi \right]\end{aligned}$$

and at P , $\mathbf{E}_P = -4(A - B) \mathbf{a}_\rho$. Thus $|\mathbf{E}_P| = \pm 4(A - B)$. Also, $V_P = A + B$. Our two equations are:

$$4(A - B) = \pm 500$$

and

$$A + B = 100$$

We thus have two pairs of values for A and B :

$$\underline{A = 112.5, B = -12.5} \text{ or } \underline{A = -12.5, B = 112.5}$$

HW-6



2. Consider a coaxial capacitor having inner radius a , outer radius b , unit length, and filled with a material with dielectric constant, ϵ_r . Compare this to a parallel-plate capacitor having plate width, w , plate separation d , filled with the same dielectric, and having unit length. Express the ratio b/a in terms of the ratio d/w , such that the two structures will store the same energy. (Please also state the reasons!)
3. Let $S = 100\text{mm}^2$, $d = 3\text{ mm}$, and $\epsilon_r = 12$ for a parallel-plate capacitor. a) Calculate the capacitance; b) After connecting a 6 V battery across the capacitor, calculate $\mathbf{E}$, $\mathbf{D}$, Q , and the total stored electrostatic energy; c) With the source still connected, the dielectric is carefully withdrawn from between the plates. With the dielectric gone, re-calculate $\mathbf{E}$, $\mathbf{D}$, Q , and the energy stored in the capacitor. d) If the charge and energy found in (c) are less than that found in (b) (which you should have discovered), what became of the missing charge and energy?
4. A parallel-plate capacitor is made using two circular plates of radius a , with the bottom plate on the xy plane, centered at the origin. The top plate is located at $z = d$, with its center on the z axis. Potential V_0 is on the top plate; the bottom plate is grounded. Dielectric having radially-dependent permittivity fills the region between plates. The permittivity is given by $\epsilon(\rho) = \epsilon_0(1 + \rho^2/a^2)$. Find: a) $\mathbf{E}$; b) $\mathbf{D}$; c) Q ; d) C .

5. Two conducting plates, each 3 by 6 cm, and three slabs of dielectric, each 1 by 3 by 6 cm, and having dielectric constants of 1, 2, and 3 are assembled into a capacitor with $d = 3$ cm. Determine the two values of capacitance obtained by the two possible methods of assembling the capacitors.
6. A potential field in free space is given in spherical coordinates as:

$$V(r) = \begin{cases} [\rho_0/(6\epsilon_0)] [3a^2 - r^2] & (r \leq a) \\ (a^3\rho_0)/(3\epsilon_0 r) & (r \geq a) \end{cases} \quad \text{where } \rho_0 \text{ and } a \text{ are constants. a) Use}$$

Poisson's equation to find the volume charge density everywhere; b) find the total charge present.